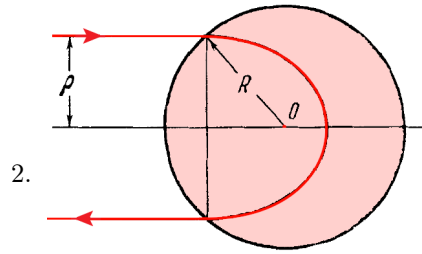


Exam №1

(Dated: Monday, 29.09.2025)

1. (**7 points**) The accelerator SuperKEKB collides electrons (e^-) of energy $E_1 = 7$ GeV with positrons (e^+) of energy $E_2 = 4$ GeV. Before the collision, both particles move in opposite directions along the axis z (collision axis in what follows). After the collision, two oppositely charged B -mesons (B^+ and B^-) are produced via the process $e^+ e^- \rightarrow B^+ + B^-$, and one of the B -mesons emerges at angle α_1 with respect to the collision axis. The rest frame masses of both B -mesons are equal to each other, whereas masses of electron and positron are negligibly small.
 - (a) (**2 points**) Find the energies of the produced B -mesons $\mathcal{E}_1, \mathcal{E}_2$ in the laboratory frame and their relative velocity v_{rel} . Your result should be expressed as a function of the angle α and energies of the incoming particles.
 - (b) (**2 points**) Assume that You can vary the energy of electrons E_1 , whereas the energy of positrons E_2 remains fixed. Find the minimal electron energy $E_{1, \text{min}}$ required for this process.
 - (c) (**3 points**) Write out explicit expressions for the Mandelstam variables s, t, u in terms of energies E_1, E_2 and the angle α in the lab frame.



The red line with arrows is the trajectory of electrons, the colored area is the region $r < R$ where the central potential $U(r)$ is not zero.

(**8 points**) The collinear beam of electrons of energy E_0 scatters in the central field $U(r)$. For the sake of simplicity, You may assume that the potential vanishes at distances $r \geq R$ from the center (point O). After the scattering, all the electrons with impact parameters $\rho < R$ are reflected back, as shown in the Figure.

- (a) (**4 points**) Find the potential $U(r)$ which satisfies these properties.
- (b) (**4 points**) A collinear beam of particles with energy $E \neq E_0$ scatters in the central potential $U(r)$ which You found earlier. Find the cross-section of elastic scattering $d\sigma/d\Omega$ for this case. Analyze what happens with the cross-section in the limit $E \rightarrow E_0$.

Solution - Exam 1

October 2, 2025

Certamen 1 - Problem 1

- ① The accelerator SuperKEKB collides electrons (e^-) of energy $E_1 = 7$ GeV with positrons (e^+) of energy $E_2 = 4$ GeV. Before the collision, both particles move in opposite directions along the axis z (collision axis in what follows). After the collision, two oppositely charged B -mesons (B^+ and B^-) are produced via the process $e^+ e^- \rightarrow B^+ + B^-$, and one of the B -mesons emerges at angle α_1 with respect to the collision axis. The rest frame masses of both B -mesons are equal to each other, whereas masses of electron and positron are negligibly small.

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Certamen 1 - Problem 1

Solution: In the lab frame, which coincides with center-of-mass frame in this problem, 4-momenta of $e^\pm$ are

$$k_{e-}^\mu = (E_1/c, \mathbf{k}_1), \quad k_{e+}^\mu = (E_2/c, \mathbf{k}_2), \quad E_{1,2} = c\sqrt{\mathbf{k}_{1,2}^2 + m_e^2 c^2} = E_e \quad (1)$$

Since masses are negligible, we may assume that $\mathbf{k}_a \approx \pm (E_a/c) \hat{\mathbf{z}}$

The “total” energy and momentum of the system are $E = E_1 + E_2$, $\mathbf{K} = \hat{\mathbf{z}} (E_1 - E_2)/c$

► After interaction we produce $B^+ B^-$ pairs, whose momenta in the center-of-mass-frame are

$$p_{B+}^\mu = (\mathcal{E}_1/c, \mathbf{p}_1), \quad p_{B-}^\mu = (\mathcal{E}_2/c, \mathbf{p}_2), \quad \mathcal{E}_a = c\sqrt{\mathbf{p}_a^2 + m_B^2 c^2} \quad (2)$$

In what follows we'll assume scattering occurs in Oxz plane and ignore the y -direction
In view of the energy-momentum conservation,

$$\mathcal{E}_1 + \mathcal{E}_2 = E_1 + E_2 = E_{\text{tot}}, \quad \mathbf{p}_1 + \mathbf{p}_2 = \hat{\mathbf{z}} (E_1 - E_2) = \hat{\mathbf{z}} \Delta E$$

If the B^+ forms angle α , and B^- forms angle β then we have (in components)

$$\sqrt{\mathcal{E}_1^2 - m_B^2} \cos \alpha + \sqrt{\mathcal{E}_2^2 - m_B^2} \cos \beta = E_1 - E_2$$

$$\sqrt{\mathcal{E}_1^2 - m_B^2} \sin \alpha + \sqrt{\mathcal{E}_2^2 - m_B^2} \sin \beta = 0$$

Certamen 1 - Problem 1

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$$\sqrt{\mathcal{E}_1^2 - m_B^2} \sin \alpha + \sqrt{\mathcal{E}_2^2 - m_B^2} \sin \beta = 0$$

Solving this system of equations in Mathematica, can get

$$\mathcal{E}_1 = \frac{2E_1E_2E_{\text{tot}} - \cos(\alpha)\Delta E \sqrt{4E_1^2E_2^2 + m_B^2 \cos^2(\alpha)\Delta E^2 - m_B^2E_{\text{tot}}^2}}{E_{\text{tot}}^2 - \cos^2(\alpha)\Delta E^2}$$

$$\mathcal{E}_2 = E_1 + E_2 - \mathcal{E}_1$$

$$\beta = \arctan \left(\frac{\sqrt{\mathcal{E}_2^2 - m_B^2} \sin \beta}{\sqrt{\mathcal{E}_2^2 - m_B^2} \cos \beta} \right) = -\arctan \left(\frac{\sqrt{\mathcal{E}_1^2 - m_B^2} \sin \alpha}{E_1 - E_2 - \sqrt{\mathcal{E}_1^2 - m_B^2} \cos \alpha} \right)$$

Certamen 1 - Problem 1

Evaluation of the relative velocity v_{rel} (consider $c = 1$ for simplicity):

- By definition, this is velocity of one of the particles in the rest frame of the other

$$p_1 = (m_1, 0), \quad p_2 = \left(\frac{m_2}{\sqrt{1 - v_{\text{rel}}^2}}, \frac{m_2 v_{\text{rel}}}{\sqrt{1 - v_{\text{rel}}^2}} \right),$$
$$\Rightarrow (p_1 \cdot p_2) = \frac{m_1 m_2}{\sqrt{1 - v_{\text{rel}}^2}} \quad (3)$$

Solving (3), can get invariant relative 4-velocity:

$$v_{\text{rel}} = \sqrt{1 - \frac{m_1^2 m_2^2}{(p_1 \cdot p_2)^2}} = \sqrt{1 - \frac{m_B^4}{(2E_1 E_2 - m_B^2)^2}}$$

and we used a chain of relation (energy momentum conservation, ...)

$$\begin{aligned} p_1 \cdot p_2 &\equiv p_{B^+} \cdot p_{B^-} = \frac{(p_{B^+} + p_{B^-})^2 - 2m_B^2}{2} = \frac{(p_{e^+} + p_{e^-})^2 - 2m_B^2}{2} \\ &= \frac{s - 2m_B^2}{2} \approx 2E_1 E_2 - m_B^2 \end{aligned}$$

Minimal energy of the process might be found from the condition $v_{\text{rel}} = 0$,

$$2E_1 E_2 - m_B^2 = m_B^2 \quad \Rightarrow \quad E_1^{(\min)} = \frac{m_B^2}{E_2}$$

Certamen 1 - Problem 1

Why do we need Mandelstam variables?

- We may study the process in different frames, e.g.:

Experiment 1: Collision of e^+e^- -beams with equal energy E_e and opposite directions

Experiment 2: Collision of e^+e^- -beams with equal energy E_e and forming angle $\pi/2$

Experiment 3: Collision of e^+e^- -beams with unequal energies $E_1 \neq E_2$ and opposite directions

Experiment 4: Collision of scattering of high-energy electron on a positron which is at rest in the lab frame

Underlying process is the same, $e^+e^- \rightarrow \pi^+\pi^-$, we just consider it in different frames.

- Transformation of kinematics (angles, energies) from one frame to another is too complicated.

Lorentz covariance:

- We'll work with Lorentz-covariant Mandelstam variables, and expect that all the invariant cross-sections depend on s, t
 - *Actually there are 3 Mandelstam variables s, t, u , yet they are bound by

$$s + t + u = \sum m_i^2,$$

so only 2 of them are independent (choose s, t in what follows)

- Need to distinguish invariant cross-sections $d\sigma/dt$ and frame-dependent

$$\frac{d\sigma}{d\theta} = \frac{d\sigma}{dt} \left(\frac{dt}{d\theta} \right)$$

Certamen 1 - Problem 1

Explicit results for Mandelstam variables (drop $m_e \approx 0$ everywhere):

$$s = (k_{e-} + k_{e+})^2 = 4E_1 E_2,$$

$$t = (k_{e-}^\mu - p_{B+}^\mu)^2 = m_B^2 - 2E_1 \mathcal{E}_1 + 2E_1 \sqrt{\mathcal{E}_1^2 - m_B^2} \cos \alpha$$

$$u = (k_{e-}^\mu - p_{B-}^\mu)^2 = m_B^2 - 2E_1 \underbrace{\mathcal{E}_2}_{E_1 + E_2 - \mathcal{E}_1} + 2E_1 \underbrace{\sqrt{\mathcal{E}_2^2 - m_B^2} \cos \beta}_{E_1 - E_2 - \sqrt{\mathcal{E}_1^2 - m_B^2} \cos \alpha}$$

and

$$\mathcal{E}_1 = \frac{2E_1 E_2 E_{\text{tot}} - \cos(\alpha) \Delta E \sqrt{4E_1^2 E_2^2 + m_B^2 \cos^2(\alpha) \Delta E^2 - m_B^2 E_{\text{tot}}^2}}{E_{\text{tot}}^2 - \cos^2(\alpha) \Delta E^2}$$

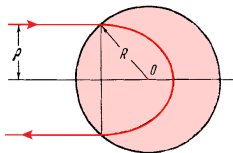
Sum of the Mandelstam variables:

$$s + t + u = 4E_1 E_2 + 2m_B^2 - 2E_1 (E_1 + E_2) + 2E_1 (E_1 - E_2) = 2m_B^2 = \text{const}$$

(all angles and $\sqrt{\dots}$ cancel in the sum, as expected)

Exam 1 - Problem 2

The collinear beam of electrons of energy E_0 scatters in the central field $U(r)$. For the sake of simplicity, You may assume that the potential vanishes at distances $r \geq R$ from the center (point O). After the scattering, all the electrons with impact parameters $\rho < R$ are reflected back, as shown in the Figure.



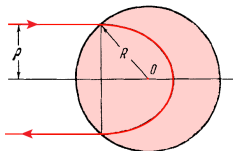
The red line with arrows is the trajectory of electrons, the colored area is the region $r < R$ where the central potential $U(r)$ is not zero.

- 1 Find the potential $U(r)$ which satisfies these properties.
- 2 A collinear beam of particles with energy $E \neq E_0$ scatters in the central potential $U(r)$ which You found earlier. Find the cross-section of elastic scattering $d\sigma/d\Omega$ for this case. Analyze what happens with the cross-section in the limit $E \rightarrow E_0$.

Goal of this problem:

Check your knowledge of inverse scattering motion in central field

Exam 1 - Problem 2



From geometric construction can see that the scattering angle

$$\theta(\rho < R) = \pi$$

and there is no scattering outside

This is classical inverse scattering problem,

– Since all particles have the same energy, can use Firsov's formula

$$w = \exp \left(\frac{1}{\pi} \int_{rw}^R \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}} d\rho \right)$$

$$\ln w = \frac{1}{\pi} \int_{rw}^R \frac{\theta(\rho)}{\sqrt{\rho^2 - r^2 w^2}} d\rho = \coth^{-1} \left(\frac{R}{\sqrt{R^2 - r^2 w^2}} \right),$$

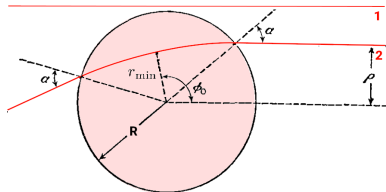
$$w = \sqrt{\frac{2R - r}{r}} = \sqrt{1 - \frac{U}{E_0}}$$

so

$$U(r < R) = 2E_0 \left(1 - \frac{R}{r} \right), \quad U(r > R) = 0$$

Exam 1 - Problem 2

Different regimes:



Should distinguish:

- For $r > R$ trajectories are just straight lines
- For $r < R$ need to find trajectories in the potential

$$U(r) = \left(2E_0 - \frac{2E_0 R}{r} \right), \quad r < R, \quad r > R$$

The first term $2E_0 = \text{const}$ just “shifts” energy, does not contribute to force $\mathbf{F} = -\nabla U$

The second term is ordinary Coulomb, need only to change $\alpha \rightarrow 2E_0 R$

$\Rightarrow$ We may literally repeat what we discussed in Kepler’s problem and get solutions as conic section

$$\frac{p}{r} = 1 + e \cos \phi, \quad p = \frac{M^2}{2mE_0 R}, \quad e = \sqrt{1 + \frac{2(E - 2E_0) M^2}{4mE_0^2 R^2}}$$

If particle’s trajectory leaves the potential well, the trajectory is a mix of straight line AND conic section

– No discontinuity at the border since $U(r)$ is continuous

Exam 1 - Problem 2

$$\frac{p}{r} = 1 + e \cos \phi, \quad p = \frac{M^2}{2mE_0 R}, \quad e = \sqrt{1 + \frac{2(E - 2E_0) M^2}{m\alpha^2 R^2}}$$

NOTE: If You use Mathematica and try to get trajectory using general expression

$$\phi - \phi_0 = \int^r dr \frac{M}{m r^2 \sqrt{\frac{2}{m} (E - U_{\text{eff}}(r))}}$$

You may get result in many equivalent forms:

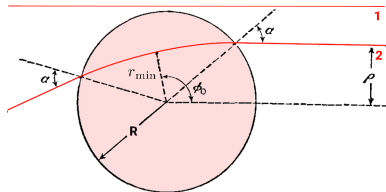
$$\phi = \arccos \left[\frac{1}{e} \left(\frac{p}{r} - 1 \right) \right]$$

$$\phi = \arcsin \left[\sqrt{1 - \frac{1}{e^2} \left(\frac{p}{r} - 1 \right)^2} \right]$$

$$\phi = \arctan \left[\frac{\sqrt{1 - \frac{1}{e^2} \left(\frac{p}{r} - 1 \right)^2}}{\frac{1}{e} \left(\frac{p}{r} - 1 \right)} \right] = \arctan \left[\frac{\sqrt{(e^2 - 1)r^2 + p^2 + 2pr}}{(p - r)} \right]$$

Exam 1 - Problem 2

Different regimes:



► for $\rho > R$ no scattering at all (!!!) $\Rightarrow \theta = 0$, but $\phi_0 = \pi/2, \neq 0$

► For $\rho < R$ we may use conventional formulas valid in any central potential (show in blue color additional terms which differ from Kepler's problem):

$$\begin{aligned}\phi_0 &= \int_{r_{\min}}^{\infty} \frac{\frac{M}{m r^2} dr}{\sqrt{\frac{2}{m} \left(E - U(r) - \frac{M^2}{2 m r^2} \right)}} = \int_{r_{\min}}^{\infty} \frac{\frac{\rho}{r^2} dr}{\sqrt{\left(1 - \frac{4E_0 \Theta(R-r)}{mv_{\infty}^2} \left(1 - \frac{R}{r} \right) - \frac{\rho^2}{r^2} \right)}} \\ &= \int_{r_{\min}}^R \frac{\frac{\rho}{r^2} dr}{\sqrt{\left(1 - \frac{4E_0}{mv_{\infty}^2} \left(1 - \frac{R}{r} \right) - \frac{\rho^2}{r^2} \right)}} + \int_R^{\infty} \frac{\frac{\rho}{r^2} dr}{\sqrt{\left(1 - \frac{\rho^2}{r^2} \right)}}\end{aligned}$$

Minimal distance $r_{\min}$ found from equation

$$1 - \frac{4E_0 U_0}{mv_{\infty}^2} \left(1 - \frac{R}{r_{\min}} \right) - \frac{\rho^2}{r_{\min}^2} = 0, \quad \rho \leq R, \quad 0 \leq r_{\min} \leq R, \quad r_{\min} = \frac{\rho}{1+e}$$

Exam 1 - Problem 2

Evaluation of the integrals in Mathematica for $\phi_0 = (\pi - \theta)/2$:

$$\phi_0 = \sec^{-1} \left(\frac{R \sqrt{m \rho^2 v^2 (m v^2 - 4 E_0) + \alpha^2 R^2}}{m \rho^2 v^2 - 2 E_0 R^2} \right) + \tan^{-1} \left(\frac{\rho}{\sqrt{R^2 - \rho^2}} \right)$$

To "join" several inverse functions, can apply known function (Sin, Cos, ...) use TrigExpand[...] and then FullSimplify[...], final results:

$$\cos \phi_0 = -2 E_0 \sqrt{\frac{R^2 - \rho^2}{m \rho^2 v_\infty^2 (m v_\infty^2 - 4 E_0) + 4 E_0^2 R^2}}, \quad \rho = \frac{2 E_0 R \sin(\phi_0)}{\sqrt{4 E_0^2 + m v^2 \cos^2(\phi_0) (m v^2 - 4 E_0)}}$$

$$\sin \phi_0 = \frac{\rho (m v_\infty^2 - 2 E_0)}{\sqrt{m \rho^2 v_\infty^2 (m v_\infty^2 - 4 E_0) + 4 E_0^2 R^2}}, \quad \rho = \frac{2 E_0 R \sin(\phi_0)}{\sqrt{(2 E_0 - m v^2)^2 - m v^2 \sin^2(\phi_0) (m v^2 - 4 E_0)}}$$

$$\tan \phi_0 = \frac{\rho (2 E_0 - m v^2)}{2 E_0 \sqrt{R^2 - \rho^2}}, \quad \rho = \frac{2 E_0 R \tan(\phi_0)}{\sqrt{4 E_0^2 + m^2 v^4 - 4 E_0 m v^2 + 4 E_0^2 \tan^2(\phi_0)}}$$

All the expressions for ρ in right-hand sides are equivalent to each other

To find cross-section, can use

$$\frac{d\sigma}{d\Omega} = \frac{1}{2\pi \sin \theta} 2\pi \rho \left| \frac{d\rho}{d\theta} \right| = \frac{1}{2 \sin \theta} \left| \frac{d\rho^2}{d\theta} \right| = \frac{4 E_0^2 R^2 (2 E_0 - m v^2)^2}{4 (4 E_0^2 \cos^2(\theta) + (2 E_0 - m v^2)^2 \sin^2(\theta))^2}$$

Exam 1 - Problem 2

$$U(r) = \begin{cases} 2E_0 \left(1 - \frac{R}{r}\right), & r < R \\ 0, & r \geq R \end{cases} \quad (1) \quad \rho = \frac{2E_0 R \cot(\theta/2)}{\sqrt{4E_0^2 + m^2 v^4 - 4E_0 m v^2 + 4E_0^2 \cot^2(\theta/2)}}$$

$$\frac{d\sigma}{d\Omega} = \frac{E_0^2 R^2 (2E_0 - m v^2)^2}{\left(4E_0^2 \cos^2\left(\frac{\theta}{2}\right) + (2E_0 - m v^2)^2 \sin^2\left(\frac{\theta}{2}\right)\right)^2} = \frac{E_0^2 R^2 (E_0 - E)^2}{\left(E_0^2 \cos^2\left(\frac{\theta}{2}\right) + (E_0 - E)^2 \sin^2\left(\frac{\theta}{2}\right)\right)^2}$$

Observations and cross-check:

- If $E = m v^2/2$ approaches E_0 , the cross-section vanishes (as expected)
- Our potential vanishes at large $r \rightarrow \infty$, so we don't have singularity at small scattering angles
- If we replace $E_0 \rightarrow E_0/R$ and after that consider limit $R \rightarrow \infty$, $U(r)$ turns into the Coulomb potential, and $d\sigma/d\Omega$ turns into Rutherford's formula